



OCR A-Level Computer Science (H446)

Component 01: Computer Systems

Practice Examination

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes provided.
- Answer ALL questions.
- Write your answers in the spaces provided.
- Additional paper may be used if required.

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is 40.
- This document consists of 8 pages.
- Quality of extended responses will be assessed in questions marked with an asterisk (*).

Candidate Name:	Centre Number:
Candidate Number:	Date:

TIME ALLOWED: 75 minutes

TOTAL MARKS: 40

For Examiner's Use:

Question	1	2	3	4	5	Total
Marks	/8	/7	/10	/8	/7	/40

Questions

1. A CPU (Central Processing Unit) consists of several components that work together to execute instructions.

(a) State the purpose of the Control Unit (CU) in a CPU. [1 mark]

(b) Describe the role of the Arithmetic Logic Unit (ALU) in a CPU. [2 marks]

(c) Explain the difference between the Program Counter (PC) and the Memory Address Register (MAR). [3 marks]

(d) Name and describe the purpose of TWO other CPU registers. [2 marks]

2. The performance of a CPU can be affected by various factors.

(a) Explain what is meant by CPU clock speed and how it affects performance. [2 marks]

(b) Explain what is meant by the term "core" in multi-core processors. [2 marks]

(c) Describe ONE advantage and ONE limitation of increasing the number of cores in a processor.
[3 marks]

3. The Fetch-Decode-Execute cycle is fundamental to the operation of a CPU.

(a) Describe the three stages of the Fetch-Decode-Execute cycle. [6 marks]

(b) Explain what is meant by "pipelining" in CPU architecture and how it improves performance.
[4 marks]

4. Different types of processors are designed for specific purposes.
- (a) Compare the characteristics and typical uses of CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) processors. [4 marks]
- (b) Explain why mobile devices often use ARM processors rather than x86 processors. [2 marks]
- (c) Describe TWO features of Graphics Processing Units (GPUs) that make them suitable for parallel processing tasks. [2 marks]

5. (a) Describe the purpose of cache memory in a computer system and explain why it improves system performance. [3 marks]

- (b) Explain the difference between Level 1 (L1) and Level 2 (L2) cache. [2 marks]

- (c) Describe what is meant by "cache hit" and "cache miss". [2 marks]

Grade Boundaries

Grade	A*	A	B	C	D
Marks	36-40	32-35	28-31	24-27	20-23